ETALE-LOCAL CRITERION FOR A COCONE TO EXHIBIT THE COLIMIT IN ALGEBRAIC SPACES

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 449-474 of sources/stacks-spaces-pushouts/pinned-original-source.tex; proof span is lines 476-550, the complete author proof (73 body lines).

Editorial note. Navigation stubs: 3 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 5 of queue_spaces-pushouts.json (pb=3562, stmt 449-474, proof 476-550).

Editorial note. spaces-pushouts wave-1: fresh adjudication accept (2026-10-14); six-check dispatched pool (§2 row appended 02:4x).

Editorial note. Preamble patch operations/agent-authoring/lane-s3/patch-etale-math.tex appended: the author's statement (source line 459) contains \'etale inside math mode - a typo in the upstream text which is retained verbatim; the patch maps \' to \acute{e} inside math groups only so the span compiles without touching the author text.

Editorial note. Revision r002: r001 failed at compile (author statement writes 'etale inside math mode, source line 459; engine log Missing character). Preamble patch patch-etale-math.tex added; author text untouched.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 0G2U: sources/stacks-spaces-pushouts/pinned-original-source.tex, source lines 550–550. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0BMP: sources/stacks-spaces-pushouts/pinned-original-source.tex, source lines 382–396. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 03FX: sources/stacks-spaces-pushouts/pinned-original-source.tex, source lines 550–550. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:spaces-pushouts.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *Let S be a scheme. Let B be an algebraic space over S. Let $\mathcal{I} \to (Sch/S)_{fppf}$, $i \mapsto X_i$ be a diagram of algebraic spaces over B. Let $(X, X_i \to X)$ be a cocone for the diagram in the category of algebraic spaces over B (Categories, Remark 0G2U). Assume that*

(1) *the base change functor $X_{spaces,étale} \to \lim X_{i,spaces,étale}$, sending U to $U_i = X_i \times_X U$ is an equivalence,*
(2) *given*
   (a) *B' affine and étale over B,*
   (b) *Z an affine scheme over B',*
   (c) *$U \to X \times_B B'$ an étale morphism of algebraic spaces with U affine,*
   (d) *$f_i : U_i \to Z$ a cocone over B' of the diagram $i \mapsto U_i = U \times_X X_i$,*

   there exists a unique morphism $f : U \to Z$ over B' such that f_i equals the composition $U_i \to U \to Z$.

Then $X = \operatorname{colim} X_i$ in the category of all algebraic spaces over B.

Proof. In this paragraph we reduce to the case where B is an affine scheme. Let $B' \to B$ be an étale morphism of algebraic spaces. Observe that conditions (1) and (2) are preserved if we replace B, X_i, X by B', $X_i \times_B B'$, $X \times_B B'$. Let $\{B_a \to B\}_{a \in A}$ be an étale covering with B_a affine, see Properties of Spaces, Lemma 03FX. For $a \in A$ denote X_a, $X_{a,i}$ the base changes of X and the diagram to B_a. For $a, b \in A$ denote $X_{a,b}$ and $X_{a,b,i}$ the base changes of X and the diagram to $B_a \times_B B_b$. By Lemma 0BMP it suffices to prove that $X_a = \operatorname{colim} X_{a,i}$ and $X_{a,b} = \operatorname{colim} X_{a,b,i}$. This reduces us to the case where $B = B_a$ (an affine scheme) or $B = B_a \times_B B_b$ (a separated scheme). Repeating the argument once more, we conclude that we may assume B is an affine scheme (this uses that the intersection of affine opens in a separated scheme is affine).

Assume B is an affine scheme. Let Z be an algebraic space over B. We have to show

$$\operatorname{Mor}_B(X, Z) \longrightarrow \lim \operatorname{Mor}_B(X_i, Z)$$

is a bijection.

Proof of injectivity. Let $f, g : X \to Z$ be morphisms such that the compositions $f_i, g_i : X_i \to Z$ are the same for all i. Choose an affine scheme Z' and an étale morphism $Z' \to Z$. By Properties of Spaces, Lemma 03FX we know we can cover Z by such affines. Set $U = X \times_{f,Z} Z'$ and $U' = X \times_{g,Z} Z'$ and denote $p : U \to X$ and $p' : U' \to X$ the projections. Since $f_i = g_i$ for all i, we see that

$$U_i = X_i \times_{f_i,Z} Z' = X_i \times_{g_i,Z} Z' = U'_i$$

compatible with transition morphisms. By (1) there is a unique isomorphism $\epsilon : U \to U'$ as algebraic spaces over X, i.e., with $p = p' \circ \epsilon$ which is compatible with the displayed identifications. Choose an étale covering $\{h_a : U_a \to U\}$ with U_a affine. By (2) we see that $f \circ p \circ h_a = g \circ p' \circ \epsilon \circ h_a = g \circ p \circ h_a$. Since $\{h_a : U_a \to U\}$ is an étale covering we conclude $f \circ p = g \circ p$. Since the collection of morphisms $p : U \to X$ we obtain in this manner is an étale covering, we conclude that $f = g$.

Proof of surjectivity. Let $f_i : X_i \to Z$ be an element of the right hand side of the displayed arrow in the first paragraph of the proof. It suffices to find an étale covering $\{U_c \to X\}_{c \in C}$ such that the families $f_{c,i} \in \lim_i \operatorname{Mor}_B(X_i \times_X U_c, Z)$ come from morphisms $f_c : U_c \to Z$. Namely, by the uniqueness proved above the morphisms f_c will agree on $U_c \times_X U_b$ and hence will descend to give the desired morphism $f : X \to Z$. To find our covering, we first choose an étale covering $\{g_a : Z_a \to Z\}_{a \in A}$ where each Z_a is affine. Then we let $U_{a,i} = X_i \times_{f_i,Z} Z_a$. By (1)

we find $U_{a,i} = X_i \times_X U_a$ for some algebraic spaces U_a étale over X. Then we choose étale coverings $\{U_{a,b} \to U_a\}_{b \in B_a}$ with $U_{a,b}$ affine and we consider the morphisms

$$U_{a,b,i} = X_i \times_X U_{a,b} \to X_i \times_X U_a = X_i \times_{f_i,Z} Z_a \to Z_a$$

By (2) we obtain morphisms $f_{a,b} : U_{a,b} \to Z_a$ compatible with these morphisms. Setting $C = \coprod_{a \in A} B_a$ and for $c \in C$ corresponding to $b \in B_a$ setting $U_c = U_{a,b}$ and $f_c = g_a \circ f_{a,b} : U_c \to Z$ we conclude. □